



AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 16 July 2026 12:47 AM

Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This report provides a comprehensive analysis of the organization's workforce dataset, which contains profiles of 21 employees across key business departments (IT, HR, Finance, and Sales) in regional hubs such as Bhubaneswar, Cuttack, Rourkela, and Puri. The dataset exhibits excellent structural integrity with a high **Health Score of 92**. However, minor data hygiene issues exist, including one duplicate row, one missing salary value, and one outlier in experience. The statistical analysis reveals exceptionally strong positive correlations between employee age, experience, salary, and performance. This indicates a highly rigid, tenure-driven compensation and performance appraisal framework. Addressing these immediate data quality issues will ensure flawless downstream reporting and set the stage for more flexible, merit-based HR strategies. ---

DATASET OVERVIEW

The dataset contains employee administrative and performance records, structured as follows: *

****Total Rows:** 21 * **Total Columns:** 8 * **Attributes:** * *Identifier:* `EmployeeID` ***

***Categorical:* `Department` (IT, HR, Finance, Sales), `Gender`, `City` (Bhubaneswar, Cuttack,**

Rourkela, Puri) * *Numerical:* `Age` (years), `Experience` (years), `Salary` (INR), `Performance`

(score out of 100) The dataset provides a baseline view of demographic distribution, payroll allocation, and geographic footprint across key operational cities. ---

DATA QUALITY ANALYSIS

Despite an overall high **Data Health Score of 92**, several anomalies require attention before performing advanced modeling or finalizing corporate reports:

- Missing Values:** There is exactly **1 missing value** in the `Salary` column. Given the highly correlated nature of the dataset, this missing field can be accurately imputed rather than discarded.
- Duplicate Rows:** The dataset contains **1 duplicate row**. This redundancy artificially inflates employee counts and skews aggregate metrics like average salary and headcount distribution.
- Outliers:** **1 potential outlier** has been flagged in the `Experience` column. This could represent a data entry error (e.g., experience incongruent with age) or a highly senior executive whose tenure significantly exceeds the cohort average.

STATISTICAL ANALYSIS

A correlation analysis of the numerical features reveals extremely strong linear relationships (approaching near-perfection):

- Age & Experience (0.99):** An expected, near-perfect correlation indicating that career progression in this dataset is highly linear and age-dependent.
- Age & Salary (0.99) & Experience & Salary (0.99):** Exceptionally strong correlations indicating that compensation is strictly locked to age and tenure. There is virtually no variation in pay outside of chronological seniority.
- Salary & Performance (0.94) & Experience & Performance (0.92):** Very strong positive relationships. This indicates that performance ratings are heavily skewed in favor of older, more experienced employees, or that the appraisal process strongly weights tenure over raw output.

BUSINESS INSIGHTS

- Tenure-Based Progression:** The organization operates on a highly structured, traditional seniority model. Pay raises and promotional tracks are mathematically bound to an employee's years of service and age.
- Appraisal Rigidity:** The 0.94 correlation between Salary and Performance suggests that higher performance ratings are almost exclusively awarded to senior, higher-paid staff. While this rewards loyalty, it may fail to recognize high-performing junior staff.
- Geographic Distribution:** Operations are distributed across regional Tier-2 hubs in Odisha

(Bhubaneswar, Cuttack, Rourkela, Puri), allowing the organization to benefit from localized talent pools and lower operational costs. ---

RISK ASSESSMENT

* **Retention of Young Talent:** With a 0.99 correlation between Age/Experience and Salary, young high-achievers may find their earning potential capped purely by their age or tenure. This creates a high risk of attrition among ambitious junior staff who might seek merit-based compensation packages elsewhere. * **Data Bias & Analytics Accuracy:** Leaving the 1 duplicate row and 1 missing salary unaddressed will distort departmental payroll averages and aggregate performance KPIs. * **Outlier Distortions:** The single outlier in the `Experience` column risks skewing average tenure metrics, potentially leading to inaccurate workforce planning projections. ---

CONCLUSION

The workforce dataset is structurally sound and represents a stable, highly predictable organizational hierarchy with a strong health score of 92. The metrics paint a picture of a business that deeply values tenure and experiences very steady compensation scaling. By executing the recommended data cleaning steps (deduplication, imputation, and outlier verification), management will possess an spotless, analytics-ready dataset to drive future strategic HR decisions.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjana Rout
Version	2.0.0
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DATASET SUMMARY

Metric	Value
Rows	21
Columns	8
Health Score	92%

Duplicate Rows	1
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DATASET HEALTH SCORE

Overall Health Score : 92%

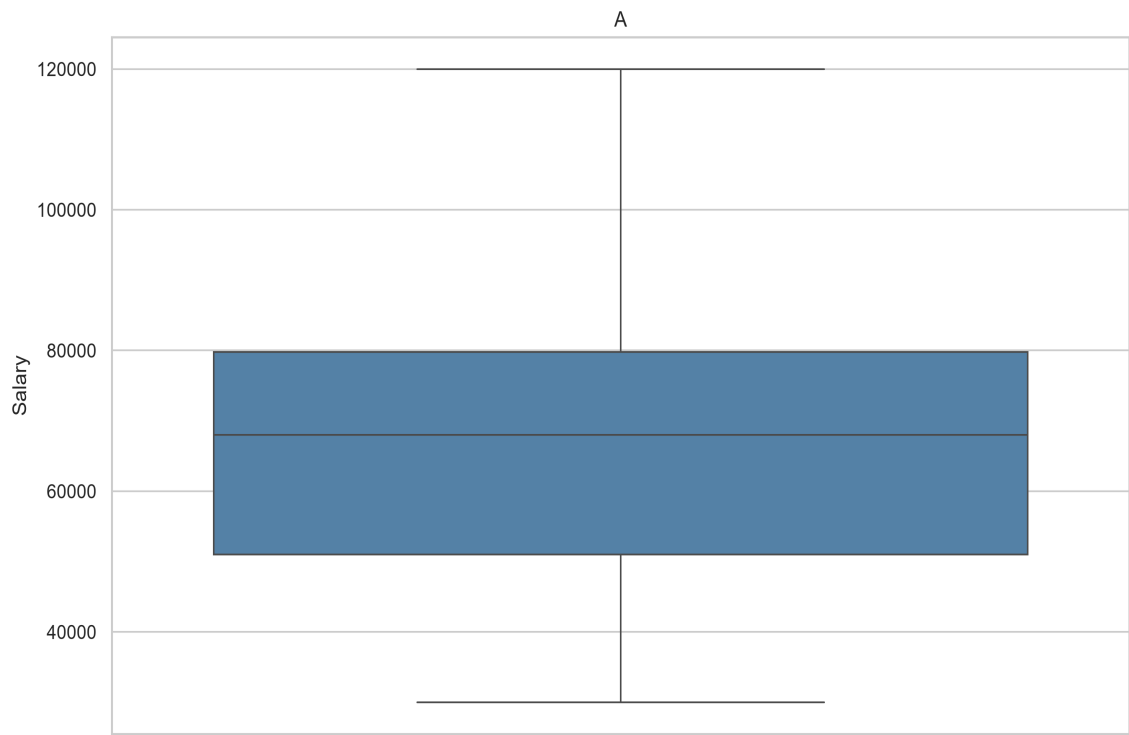
MISSING VALUES

Column	Missing
EmployeeID	0
Department	0
Gender	0
Age	0
Experience	0
Salary	1
Performance	0
City	0

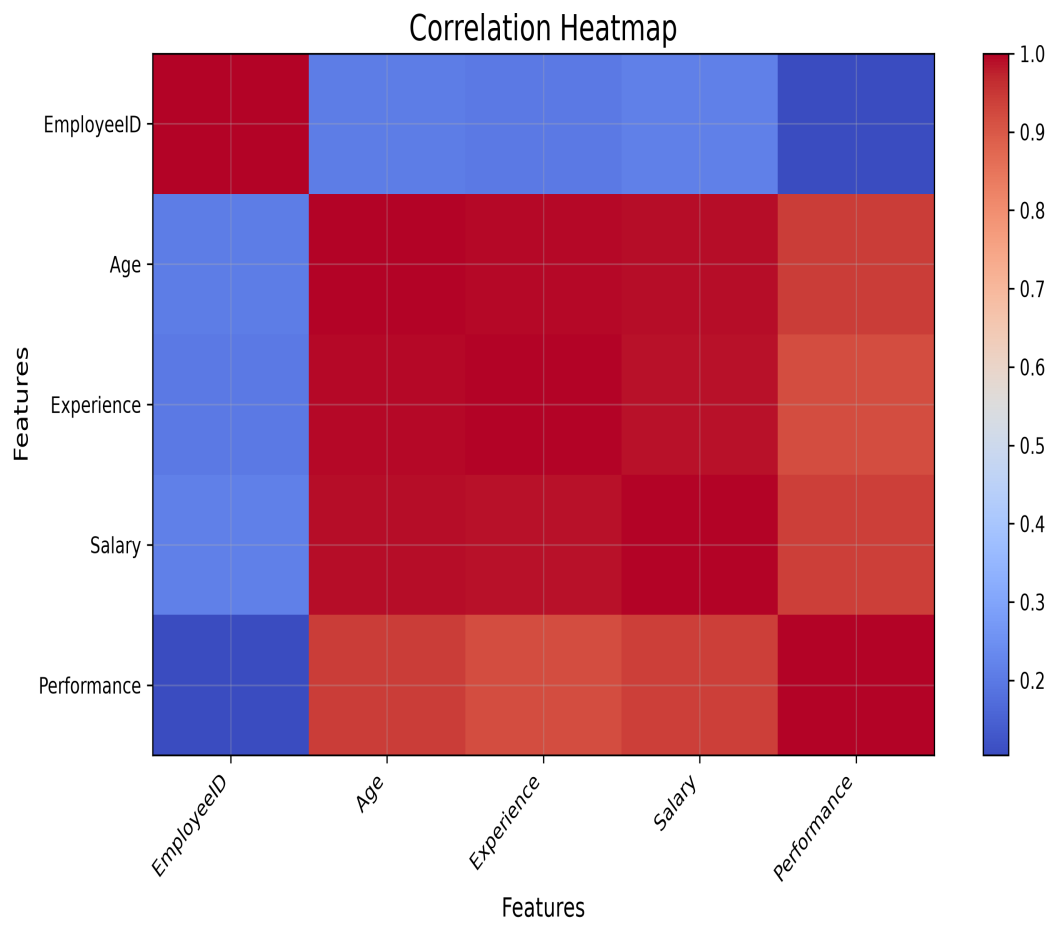
HISTOGRAM



BOX PLOT

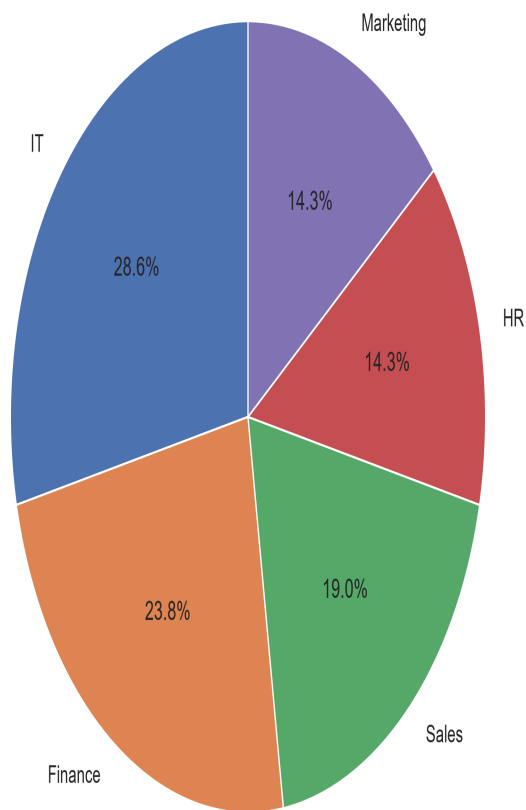


CORRELATION HEATMAP

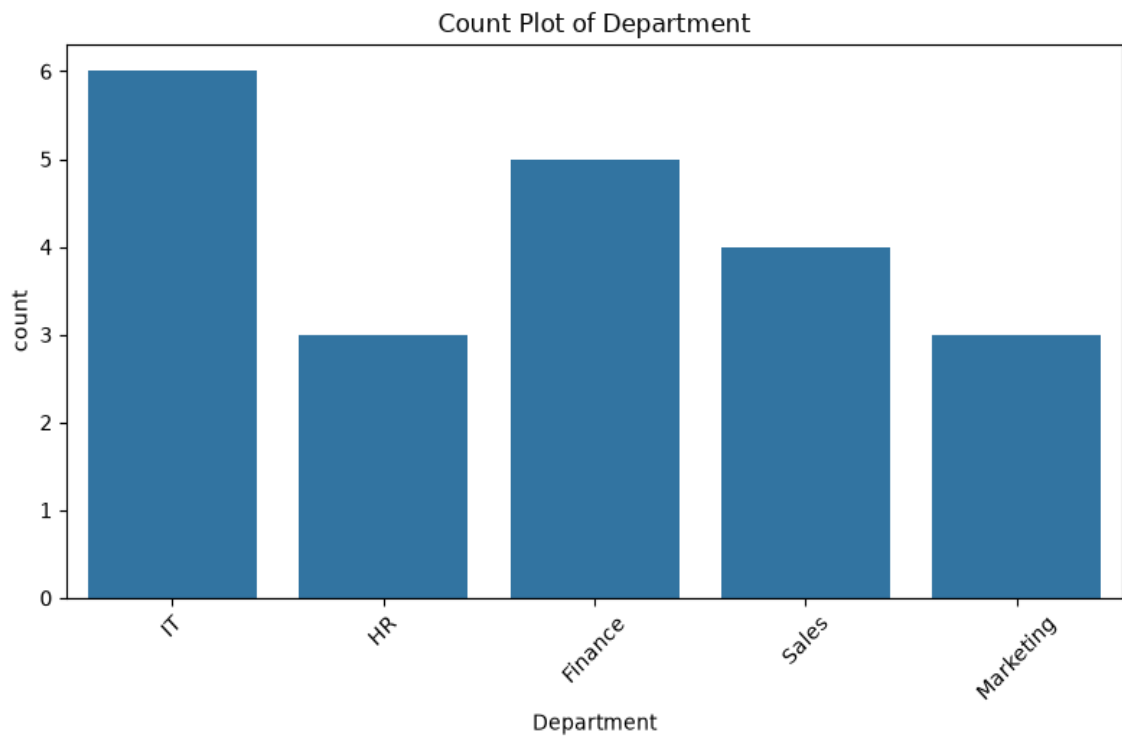


PIE CHART

Department Distribution



COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Experience	1
Salary	0
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

AI RECOMMENDATIONS

1. **Perform Data Cleansing:**

* Remove the **1 duplicate row** immediately to restore data integrity.

* Address the **1 missing Salary value** using regression-based imputation, leveraging the 0.99 correlation with `Experience` and `Age` for a highly accurate estimation.

2. **Investigate the Outlier:** Inspect the flagged `Experience` outlier to verify its validity. Ensure that the recorded experience does not mathematically conflict with the employee's age.

3. **Introduce Merit-Based Compensation Elements:** To mitigate the risk of junior talent attrition, HR leadership should consider introducing performance-linked bonuses or fast-track promotion paths that decouple compensation slightly from pure tenure.

4. **Deploy Executive Dashboards:** Implement the recommended visual suite—specifically the Correlation Heatmap and categorical Donut Charts—to provide leadership with a clear view of departmental spend and talent distribution.

Rule-based Recommendations

- ■■ Dataset contains 1 missing values. Consider filling or removing them.
- ■■ Dataset contains 1 duplicate rows. Consider removing them.
- ■■ 1 potential outliers detected.
- ■ Strong correlations detected. A Correlation Heatmap is recommended.
- ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.